INTERNAL TECHNICAL REFERENCE

ZenoHosp HMS IPD & Medical Records Schema Reference

Database schema, enum values, and timeline assembly for the IPD log, clinical records, and prescriptions modules.

PROJECT	DATE	SCOPE
ZenoHosp HMS	June 2026	Read-only inventory

1 IPD Log & Medical Records Tables

Clinical Records `patient_records`

The single polymorphic table for every clinical entry during a patient's stay. Record type is discriminated by `history_type_id` (integer FK). A `PRESCRIPTION` -type record additionally owns child rows in `prescription_items`.

COLUMN	TYPE	NOTES
history_id	UUID PK	Primary key
hospital_id	UUID FK	→ hospitals
patient_id	UUID FK	→ patients
created_by	UUID FK	→ users (authoring doctor/staff)
history_type_id	INTEGER FK	Converted by HistoryTypeConverter — see enum values below
description	TEXT	Clinical narrative / free-text notes
instructions	TEXT	Patient-facing care instructions (separate from description)
next_visit_date	TIMESTAMP	Generic follow-up field — not actively used for IPD
admission_id	UUID	Scopes record to an IPD admission (nullable for OPD)
admission_number	VARCHAR(50)	Denormalised admission number
appointment_id	UUID	OPD audit trail, nullable
mrn	VARCHAR(30) UNIQUE	Medical Record Number
created_at	TIMESTAMP	Immutable — set on insert only

history_type_id — enum values

Persisted as integer via `HistoryTypeConverter.java`. Reference table: `record_history_types`.

ID	CODE	DESCRIPTION
1	CONSULTATION	Doctor examination / specialist consult note
2	PRESCRIPTION	Drug prescription — has child rows in prescription_items
3	LAB_RESULT	Laboratory investigation results
4	SURGERY	OT / surgery / procedure notes
5	DIAGNOSIS	Diagnostic statement
6	OTHERS	Miscellaneous / uncategorised note
7	PROGRESS_NOTE	Daily ward round / progress note (doctor)

CONSULTATION PRESCRIPTION LAB_RESULT SURGERY DIAGNOSIS OTHERS PROGRESS_NOTE

IPD Admissions `admissions`

Core IPD admission record. Tracks the patient's full stay — room/bed assignment, attender, status, and discharge linkage.

COLUMN	TYPE	NOTES
id	UUID PK	
hospital_id	UUID FK	
patient_id	UUID FK	
room_id	UUID FK nullable	Current room assignment
bed_id	UUID FK nullable	Current bed assignment
previous_room_id	UUID FK nullable	Tracks room transfer history
admitting_doctor_id	UUID FK nullable	
department_id	UUID FK nullable	
source_appointment_id	UUID FK nullable	OPD appointment that triggered admission
admission_number	VARCHAR(30)	Human-readable admission number
ipd_id	VARCHAR(20) UNIQUE	Short IPD identifier
admission_date	TIMESTAMP NOT NULL	
actual_discharge_date	TIMESTAMP	Set when status → DISCHARGED
approx_discharge_date	TIMESTAMP	Estimated discharge (clinical planning)
admission_type_id	INTEGER FK	→ OPD_REFERRAL, EMERGENCY, DIRECT
admission_source_id	INTEGER FK	→ OPD_REFERRAL, EMERGENCY_WALK_IN, DIRECT, TRANSFER_IN
chief_complaint	TEXT	
primary_diagnosis	TEXT	
attender_name	VARCHAR(100)	
attender_phone	VARCHAR(20)	
attender_relationship	VARCHAR(50)	
status_id	INTEGER FK	→ ADMITTED, DISCHARGED, TRANSFERRED, ABSCONDED
version	BIGINT	Optimistic locking (JPA @Version)
created_by	UUID FK	
created_at / updated_at	TIMESTAMP	Audit timestamps

Discharge Details `discharges`

OneToOne with `admissions`. Populated when the admission's status flips to DISCHARGED. Holds the clinical and follow-up context of discharge.

COLUMN	TYPE	NOTES
`id`	`BIGINT PK`	Auto-increment
`admission_id`	`UUID FK UNIQUE`	OneToOne → admissions
`actual_discharge_date`	`TIMESTAMP`	
`discharge_diagnosis`	`TEXT`	Final diagnosis on discharge
`discharge_note`	`TEXT`	Free-text discharge summary note
`follow_up_date`	`DATE`	Scheduled follow-up OPD date
`follow_up_doctor_id`	`UUID nullable`	
`discharged_by`	`VARCHAR(150)`	Name of discharging doctor/staff
`discharged_at`	`TIMESTAMP`	

2 Prescriptions & Drug Lines

Design pattern

A prescription is a `patient_records` row with `history_type_id = 2` (PRESCRIPTION). Each drug line is a separate row in `prescription_items` linked by `history_id`. Deleting the parent record cascades to all drug lines.

Drug Lines `prescription_items`

One row per prescribed drug. Parent must be a PRESCRIPTION-type `patient_records` row. The pharmacy reads these rows to fulfil dispense requests.

COLUMN	TYPE	NOTES
id	UUID PK	
history_id	UUID FK	→ patient_records (CASCADE DELETE)
drug_id	UUID FK nullable	→ pharmacy_drug_master (soft link)
drug_name	VARCHAR(255) NOT NULL	Denormalised drug name
drug_generic	VARCHAR(255)	Denormalised generic name
drug_strength	VARCHAR(100)	e.g. "500mg", "10mg/5ml"
drug_form	VARCHAR(50)	TAB, CAP, SYRUP, INJ, DROPS, OINT, INH, ...
dose	VARCHAR(100)	Per-administration amount ("1 tablet", "10ml")
frequency_id	INTEGER FK	→ prescription_frequencies (see below)
duration_days	INTEGER nullable	Course length in days
quantity	INTEGER NOT NULL	Total units to dispense
route_id	INTEGER FK	→ prescription_routes (see below)
instructions	TEXT	Per-drug notes ("after meals", "with milk", "taper")
display_order	INTEGER	Ordering on printed prescription
dispensed_qty	INTEGER DEFAULT 0	Running pharmacy tally
dispense_status	VARCHAR(16)	PENDING / PARTIAL / DISPENSED
created_at	TIMESTAMP	Immutable

prescription_frequencies reference values

OD BD TDS QID Q4H Q6H Q8H HS AC PC SOS STAT

prescription_routes reference values

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3 Vitals, Nursing Notes, MAR & Orders

⚠ **Only one vitals-related table exists — and it is OPD-scoped only.**
No tables exist for IPD longitudinal vitals, nursing notes, nursing shift records, medication administration records (MAR), or doctor orders. These are the primary gaps relative to real Indian ward documentation practice.

OPD Visit Vitals `appointment_vitals`

Captures pre-consultation vitals per OPD appointment. **One row per appointment** (unique constraint on `appointment_id`). Not usable for IPD longitudinal vitals charting (e.g. every-shift BP/SpO_2 recordings).

COLUMN	TYPE	NOTES
id	UUID PK	
appointment_id	UUID UNIQUE NOT NULL	One row per OPD appointment
hospital_id	UUID NOT NULL	
patient_id	INTEGER NOT NULL	
bp_systolic	INTEGER	mmHg
bp_diastolic	INTEGER	mmHg
spo2	INTEGER	% (0–100)
heart_rate	INTEGER	Beats per minute
weight_kg	NUMERIC(5,2)	Up to 999.99 kg
height_cm	INTEGER	
blood_glucose	INTEGER	
recorded_by	UUID FK	→ users (nurse who took reading)
recorded_at	TIMESTAMP	
updated_at	TIMESTAMP	Nurse re-take overwrites same row

Tables searched for — not found

CONCEPT	STATUS	WHAT THIS MEANS
IPD Vitals (longitudinal)	❌ Does not exist	No per-shift BP / SpO_2 / temp tracking for admitted patients
Nursing Notes	❌ Does not exist	No nurse-authored narrative documentation
Nursing Shift Records	❌ Does not exist	No shift handover or assignment tracking

CONCEPT	STATUS	WHAT THIS MEANS
Medication Administration Record (MAR)	❌ Does not exist	dispensed_qty in prescription_items tracks pharmacy dispense, not nurse administration
Doctor Orders	❌ Does not exist	No structured order entry — orders are embedded in prescription / consultation free text

4 IPD Log Timeline Assembly

No single timeline endpoint exists.
The timeline is assembled entirely client-side in `IPDDetailPane.jsx` by making 5–6 parallel API calls and merging their results into one sorted event array.

fetchLogs() — parallel API calls

SOURCE	ENDPOINT	EVENT TYPES PRODUCED
`Hardcoded`	`(from admission object)`	ADMITTED event, DISCHARGED event
`Clinical records`	`GET /records/patient/{id}?admissionId=…`	All PatientRecord types (PROGRESS_NOTE, CONSULTATION, PRESCRIPTION, LAB_RESULT, SURGERY, DIAGNOSIS, OTHERS)
`Radiology`	`GET /radiology/admission/{admissionId}`	Order status changes (PENDING_SCAN → AWAITING_REPORT → REPORT_GENERATED)
`Room logs`	`GET /rooms/{roomId}/logs`	ALLOCATED, DEALLOCATED, ATTENDER_ASSIGNED, ATTENDER_UPDATED
`Ambulance`	`GET /ambulance/bookings`	Ambulance dispatch events (filtered by patient)
`OT bookings`	`GET /api/ot/bookings`	Surgical procedure events (filtered by admissionId)

Merge logic

All events are pushed into a single `events[]` array as plain objects with a common shape:

```
// common event shape used across all sources
{
  id:          "rec-{uuid}" | "room-{id}" | "rad-{id}" | …,
  type:        "RECORD" | "ROOM" | "RADIOLOGY" | "AMBULANCE" | "OT",
  title:       "Progress Note" | "Room Allocated" | …,
  subtitle:    "MRN-001 · Dr. Arjun",
  description: "…free text…",
  timestamp:   Date,
  // optional extra fields per source:
  badge:       "IPD",
  tone:        "info" | "warning" | "success" | …,
}
```

The merged array is sorted by `timestamp` descending (most recent first) and rendered as a vertical timeline rail in `LogTab`.

5 Complete Reference Table Seed Values

All seeded by `DataSeeder.java` via idempotent `INSERT … ON CONFLICT DO NOTHING`. Schema for every reference table: `(id INTEGER PRIMARY KEY, code VARCHAR(50))`.

admission_statuses

`ADMITTED` `DISCHARGED` `TRANSFERRED` `ABSCONDED`

admission_types

`OPD_REFERRAL` `EMERGENCY` `DIRECT`

admission_sources

`OPD_REFERRAL` `EMERGENCY_WALK_IN` `DIRECT` `TRANSFER_IN`

6 Gap Analysis — Indian Ward Documentation

WARD DOCUMENTATION NEED	STATUS	HOW IT'S CURRENTLY HANDLED
`Daily progress notes (doctor)`	✅ Exists	patient_records with PROGRESS_NOTE type (id 7, added 2026-06-08)
`Specialist consultation notes`	✅ Exists	patient_records with CONSULTATION type
`OT / surgery notes`	✅ Exists	patient_records with SURGERY type + OT bookings timeline event
`Prescription / drug order`	✅ Exists	patient_records (PRESCRIPTION) + prescription_items
`Lab investigation results`	✅ Exists	patient_records with LAB_RESULT type + radiology orders
`Diagnosis statement`	✅ Exists	patient_records with DIAGNOSIS type + primary_diagnosis on admissions
`Discharge summary`	✅ Exists	discharges table + PrintDischargeSummary page
`Vitals per shift (nurse, every 4-6h)`	❌ Missing for IPD	appointment_vitals is OPD-only (one row per visit, not per shift)
`Nursing notes / shift handover`	❌ Missing	No table exists
`Medication administration (MAR)`	❌ Missing	dispensed_qty tracks pharmacy dispense only, not nurse administration
`Doctor verbal / written orders`	❌ Missing	Embedded in prescription / consultation free text only

🏥 **Medico-legal note**

Vitals charting and nursing notes are both high-frequency and medico-legally significant in Indian hospital practice. Their absence means the current system cannot produce a complete bed-head ticket or nursing care record for regulatory or insurance audit purposes.